Formosa Telecom: listen to what customers tell support

DeployAI 2026 · Commercial proposal for Vivian Hsu, CMO · data to 30 June 2026

1. Executive summary

> **The finding that changes the plan.** When a customer tells a support agent they are thinking of leaving, about half of them are gone within 60 days (49.3% vs 3.7% for all postpaid customers, tested on three past dates using only notes written before each date). Formosa's current renewal push, aimed at contracts ending within four months, reaches people who leave at 4.2%: barely above average. Nobody reads these notes today, and 638 of them were not even tagged as a cancellation risk.

NT$2.1M–NT$7.4M	**NT$2.1M–NT$2.5M**	**NT$1.4M–NT$4.2M**
a year, keep customers who say they may leave	a year, sell only what customers ask for	a year, move paid spend to the cheapest channel

Net margin at Formosa's full scale (sample ×130), 60% margin, six months of credited revenue per customer as Finance counts it. Ranges and assumptions in Part 4.

1 · Reduce churn

Do: read every support note each day, flag postpaid customers who say they may leave, and answer their reason with the approved offer that fits it (Renewal discount, Data boost or Device credit), calling the highest-bill customers first. Billing errors and network problems get fixed, not discounted. **Worth:** NT$2.1M to NT$7.4M net margin a year, keeping 2,566–8,981 customers.

2 · Earn more from existing customers

Do: stop roaming blasts: against their control group they caused no extra sales (NT$2.0M a year). Broad upsell blasts report 3.5× but really return 1.6×, below break-even on margin. Send upgrade, data add-on and roaming offers only to customers who asked for them in a note. **Worth:** NT$2.1M to NT$2.5M a year, mostly from spend no longer wasted.

3 · Win new customers

Do: shift paid budget toward Google Search (cheapest line, NT$1,499) and away from Retail Promotion (NT$1,610); every paid channel takes about six months to pay back, and 2026 campaigns pay back slower than 2025. Add a welcome check-in: customers in their first six months leave at 9.5% in 60 days, 2.6× the average. **Worth:** NT$1.4M to NT$4.2M more margin a year at NT$20–60M of paid spend.

> **The one decision:** approve a 90-day retention pilot for postpaid customers who tell support they may leave. Expected cost NT$1.1M (at most NT$1.9M, inside the NT$2.5M budget). One in five is held back at random to prove the result. It pays for itself if it keeps 2.2% of would-be leavers, and can prove a save rate of 6.2% or more. Paid offers stop at day 45 if contacted customers leave as often as held-back ones.

2. Data exploration and cleaning

> The files were usable after 22 fixes. The ones that would have changed answers: **250 duplicated customers**, **162 duplicated campaign contacts**, **40 duplicated tickets**, contract end dates written three ways (2026-01-31, 31/01/2026 and 2026年1月31日; 13828 rows), and **6,394 contacts whose 30-day tracking window had not finished** by 30 June, which would have made recent campaigns look worse. Agent tags are unreliable: 638 notes describe a customer about to leave but carry another tag.

File	Fix	Rows before	Rows after	Detail
subscribers.csv	strip spaces in plan_type	50,250	50,250	504 values had extra spaces
subscribers.csv	strip spaces in region	50,250	50,250	500 values had extra spaces
subscribers.csv	strip spaces in device_tier	50,250	50,250	501 values had extra spaces
subscribers.csv	drop exact duplicate rows	50,250	50,000	
subscribers.csv	unify contract_end date formats	50,000	50,000	13828 dates were dd/mm/yyyy or 年月日 format
subscribers.csv	flag data_gb 999 as unlimited; keep missing as unknown	50,000	50,000	13755 unlimited, 750 missing
subscribers.csv	fill missing region with 'unknown'	50,000	50,000	750 missing
subscribers.csv	fill missing device_tier with 'unknown'	50,000	50,000	750 missing
subscribers.csv	flag prepaid bills under NT$100 as suspect (kept)	50,000	50,000	20 rows
subscribers.csv	check churn_flag vs churn_date	50,000	50,000	0 churned rows without a date
support_tickets.csv	strip spaces in category	8,040	8,040	80 values had extra spaces
support_tickets.csv	drop exact duplicate rows	8,040	8,000	
support_tickets.csv	unify date formats; missing dates kept but excluded from time tests	8,000	8,000	120 tickets have no date
support_tickets.csv	fill missing category with 'untagged'	8,000	8,000	120 missing
support_tickets.csv	notes with no text kept, read as 'no note'	8,000	8,000	120 empty notes
support_tickets.csv	check sub_id joins to subscribers	8,000	8,000	0 tickets without a customer
campaigns.csv	strip spaces in channel	240	240	2 values had extra spaces
campaigns.csv	strip spaces in target_segment	240	240	2 values had extra spaces
campaigns.csv	check spend = media + messaging + offer + creative	240	240	0 mismatches
campaigns.csv	fill missing target_segment	240	240	4 missing
campaign_contacts.csv	drop exact duplicate rows	107,849	107,687	
campaign_contacts.csv	flag contacts after 31 May 2026 (30-day window not finished)	107,687	107,687	6394 rows flagged, excluded from lift measurement

Values we could not trust, and what we did

- **Latest values only.** Plan, contract end and tenure describe 1 July (or the last day), not the past. We never use contract end or plan to pick customers in the backtest except for the comparison rule, and tenure is recounted to each test date.
- **data_gb = 999** means unlimited; it is a flag, not 999 GB. 750 customers have no allowance recorded and are treated as unknown.
- **Vague campaign targets.** 42 past campaigns were briefed as "TBD", "everyone" or "not yet defined".
- **Checks that keep problems out:** every run re-counts rows at each step; any note sentence the AI has not read sends that note to a review queue as "unknown" instead of guessing.

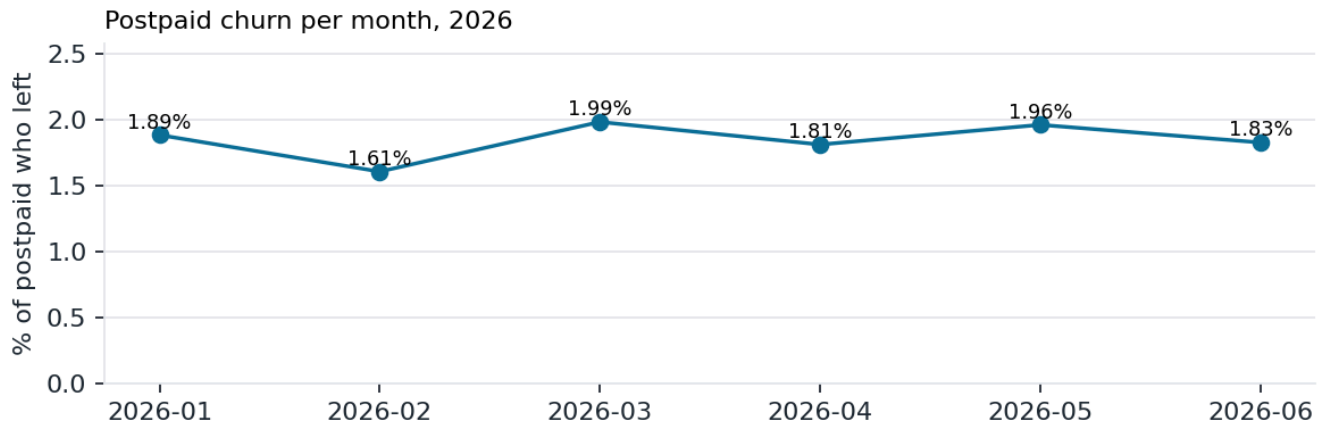
3. Analysis

> Postpaid churn runs at about 1.8% a month, as the fact sheet says. What customers tell support is by far the strongest early warning in the data: customers whose notes show leave intent leave 13× more often than average. New customers (first six months) are the second warning. Price and rival offers are the main reason flagged customers give. Past campaigns overstate their effect: once compared with the control group, roaming adds nothing and upsell returns less than half of what was reported.

The customer base

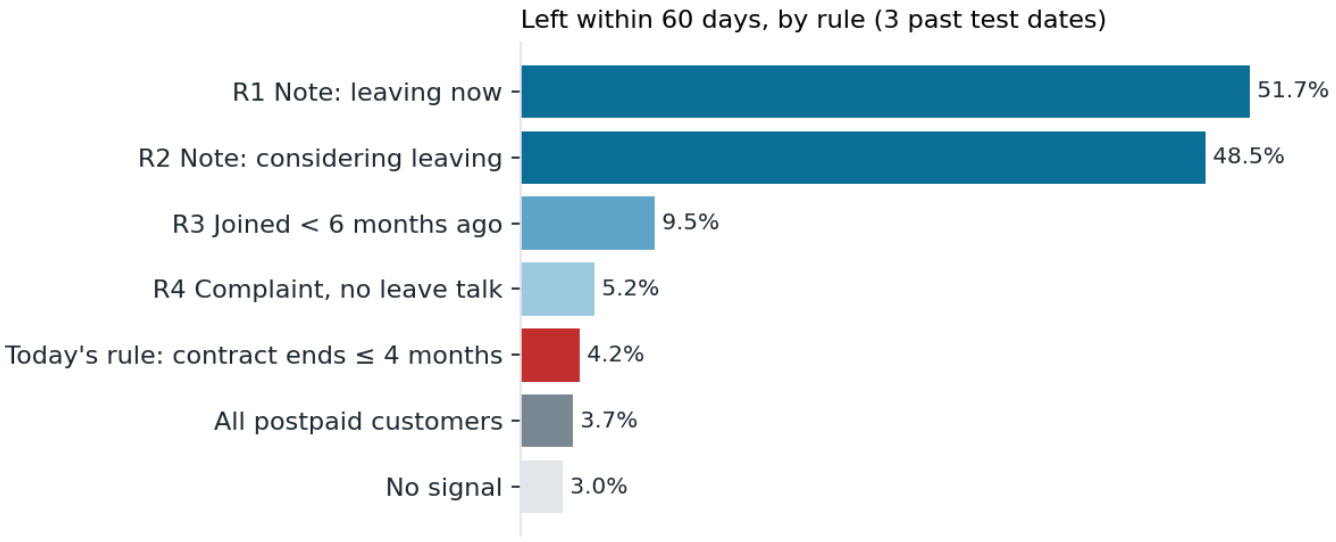
Plan	Customers (sample)	Customers (full base)	Average bill	Left Jan–Jun 2026
Prepaid	14,879	1,934,270	NT$280	17.5%
Postpaid basic	15,026	1,953,380	NT$551	11.4%
Postpaid premium	10,148	1,319,240	NT$950	8.9%
Family (other team)	5,873	763,490	NT$1,398	8.0%
Business (other team)	4,074	529,620	NT$2,202	7.8%

Family and business customers belong to other teams: 4,680 of them show leave intent in notes and are listed in flagged_family_business.csv for those teams, never targeted.



Early warning signs: tested on the past

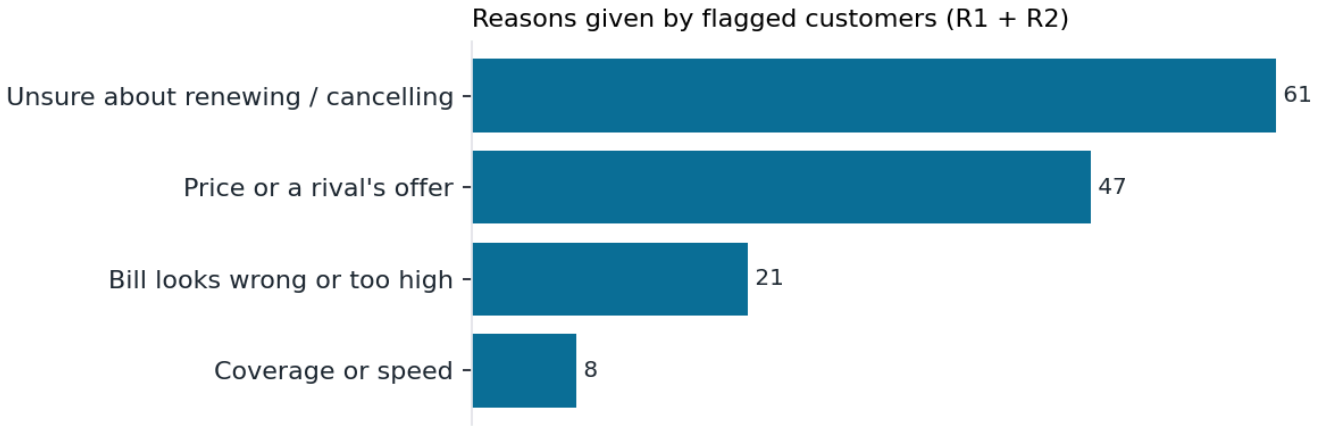
We pretended it was 1 March, 1 April and 1 May 2026, read only the notes written in the 90 days before that day, put every postpaid customer into one tier, and checked who left in the next 60 days.



Rule	Customers (3 tests)	Left in 60 days	Churn rate	× average	Share of all leavers
R1 · note: leaving now	118	61	51.7%	14.1×	2.4%
R2 · note: considering leaving	342	166	48.5%	13.2×	6.4%
R3 · joined < 6 months ago	3,343	319	9.5%	2.6×	12.4%
R4 · complaint, no leave talk	995	52	5.2%	1.4×	2.0%
No signal	65,438	1,977	3.0%	0.8×	76.6%
Excluded: line closing, no save	62	6	9.7%	2.6×	0.2%
Today's rule: contract ends ≤ 4 months	16,815	705	4.2%	1.1×	27.3%
All postpaid customers	70,298	2,581	3.7%	1.0×	100.0%

The note tiers are small but sharp: about 1 in 2 leave. They catch 8.8% of all leavers, because most leavers never call support. R3 widens the net at lower precision.

Why they leave



What customers say about rivals

Notes name a rival 417 times. Customers who name one left at 53%–62% in January–June, against 12% for all customers. More than a third are already acting (a quote, a store visit, a porting request), so speed matters more than the size of the offer. Where the rival simply offers more for the money, Data boost (NT$300) is the cheapest approved answer.

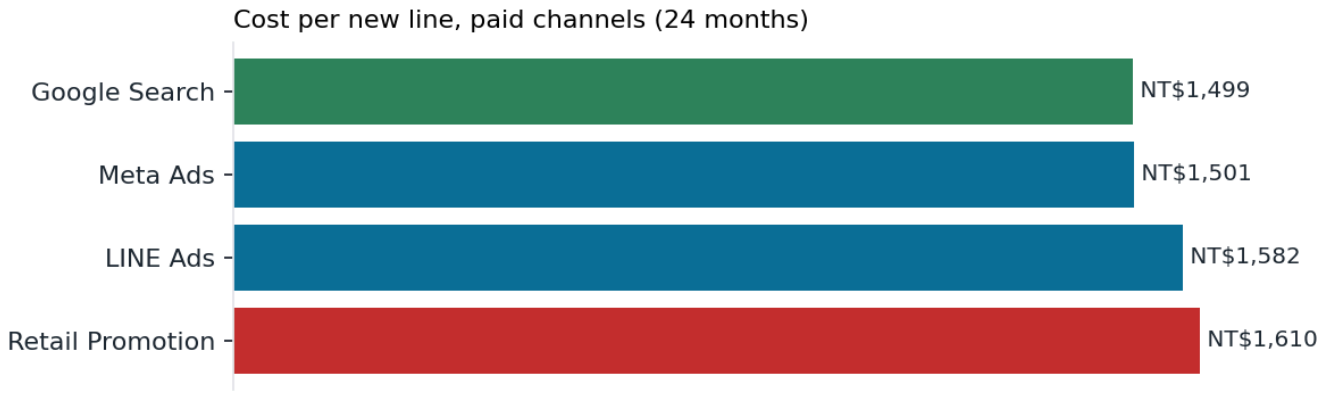
What the note says	Sentences	Share
Already acting: quote, store visit, porting, signing	150	36.0%
Rival gives more for the money	125	30.0%
Friends, family or colleagues switched	93	22.3%
Considering after seeing an offer	49	11.8%

Rival named	Customers (sample)	Mentions	Left Jan–Jun 2026
Skyward Mobile	149	151	53.0%
PacificTel	142	142	57.0%
IslandCom	121	124	62.0%

Room to earn more

22,490 customers at full scale asked about more data, a bigger plan or roaming in a note in the last 90 days. That is a short, warm list; past broad blasts went to tens of thousands with no reason to buy. Not every offer earns money: the first Roaming pass is free (NT$499 cost) and each later pass adds NT$299 of margin, so a buyer must take two more passes within six months before it pays. Plan upgrade earns about NT$1,000 margin per buyer and goes first. 38 prepaid customers in the sample (about 4,940 at full scale) asked to move to a monthly plan. There is no approved offer for that today; we flag it as unmet demand.

How acquisition performs



Channel	Campaigns	Spend	New lines	Cost per line	6-month margin return	Payback (months)
Google Search	22	NT$17.3M	11,517	NT$1,499	1.06×	5.7
Meta Ads	37	NT$31.7M	21,124	NT$1,501	0.98×	6.1
LINE Ads	23	NT$19.1M	12,076	NT$1,582	0.98×	6.3
Retail Promotion	22	NT$17.7M	11,003	NT$1,610	0.93×	6.6

Do the new customers stay?

We followed the new lines each paid channel brought in (sample) to 1 July 2026. Between 72% and 75% were still customers, so no channel brings in clearly worse customers. Counting only customers who stayed, Google Search is still the cheapest. The sample is small (about 90–160 lines per channel), so the gaps between channels are indicative only.

Channel	New lines (sample)	Still a customer 1 Jul	Cost per new line	Cost per customer who stayed
Google Search	88	75.0%	NT$1,499	NT$1,999
Meta Ads	162	74.7%	NT$1,501	NT$2,009
LINE Ads	92	71.7%	NT$1,582	NT$2,205
Retail Promotion	86	72.1%	NT$1,610	NT$2,233

4. Interpretation: what it is worth

> At Formosa's full scale the retention play is worth NT$2.1M–NT$7.4M net margin a year, the earn-more play NT$2.1M–NT$2.5M, and the acquisition shift NT$1.4M–NT$4.2M. The number that would change our mind is the save rate: nobody knows it yet, which is why the pilot measures it first.

1 · Churn

Case	Save rate	Flagged / month	Kept / year	Credited revenue	Offer cost	Net margin
low	10.0%	4,333	2,566	NT$7.8M	NT$2.5M	NT$2.1M
mid	20.0%	4,333	5,132	NT$15.5M	NT$5.1M	NT$4.2M
high	35.0%	4,333	8,981	NT$27.1M	NT$8.9M	NT$7.4M

2 · Earn more

Case	Take rate	Audience / month	Buyers / year	Credited revenue	Offer cost	Net margin
low	3.0%	7,497	2,699	NT$2.2M	NT$1.2M	NT$152K
mid	6.0%	7,497	5,398	NT$4.4M	NT$2.3M	NT$304K
high	10.0%	7,497	8,996	NT$7.4M	NT$3.9M	NT$506K

Plus NT$2.0M a year of roaming campaign spend that caused no extra sales. The earn-more range in Part 1 adds this saving.

3 · Win new

Paid spend / year	Lines at Google Search cost	Lines at Retail cost	Extra lines	Extra 6-month margin
NT$20.0M	13,342	12,421	921	NT$1.4M
NT$40.0M	26,683	24,842	1,841	NT$2.8M
NT$60.0M	40,025	37,263	2,762	NT$4.2M

Upper bound: not all spend can move to one channel. Counted on Finance's six-month rule.

Beside the six-month credit: a 24-month view

Finance books six months. Customers who stay longer are worth more: at the average postpaid churn of 1.8% a month, a new line pays about 20 of the next 24 months.

Channel	Cost per new line	Margin, 6 months	Return, 6 months	Margin, 24 months	Return, 24 months
Google Search	NT$1,499	NT$1,595	1.06×	NT$5,190	3.46×
Meta Ads	NT$1,501	NT$1,476	0.98×	NT$4,805	3.20×
LINE Ads	NT$1,582	NT$1,552	0.98×	NT$5,050	3.19×
Retail Promotion	NT$1,610	NT$1,492	0.93×	NT$4,855	3.02×

The assumptions that matter most

Assumption	Value	Why

Share of would-be leavers we keep after contact	20% planning figure; 10% and 35% shown	Unknown (Vivian, round 2). Why 20%: about 4 in 5 reasons flagged customers give (price, rival, renewal) have a matching approved offer; we assume one in four of those would-be leavers accepts and stays. Below 2.2% the pilot loses money; 2.2%-6.2% makes money but cannot be proven in 90 days; 6.2%+ is proven.
Share of flagged customers who leave within 60 days without action	from backtest	Measured on Mar/Apr/May 2026 cutoffs, notes before cutoff only.
Customers who take the offer per customer saved	1.5	Some customers who would have stayed also take the offer; this cost is counted.
Credit for one saved customer	monthly bill x months they keep paying (max 6)	Finance rule, confirmed by Vivian: six months of the full bill, stopping if they cancel. Flagged customers who are still with us after 60 days leave at about 12% a month (data), so a kept customer pays about 4.4 of the 6 months on average.
Outbound call cost	NT$2 per number dialled + NT$25 per conversation	Vivian: two charges, per conversation not per minute. Solved exactly from the 6 call campaigns in campaigns.csv; about 7.5% of dials became conversations. Care call-backs count toward the 2,000 a month.
Longer view, shown beside the 6-month credit	24 months at the observed churn rate	Vivian: start with the six-month credit, a lifetime view may sit beside it. New customers: 1.8% monthly churn (postpaid average). Kept at-risk customers: their own higher rate.
Share of customers who asked about it in a note and then buy	3% / 6% / 10%	Past broad upsell added about 0.16 points over holdout; asked-for offers should do better. Pilot measures it.
Sample to full base	x130	FACT_SHEET: sample is about 1 in 130.
New flagged customers per month	90-day audience / 3	Audience uses notes from the last 90 days.

What would change our mind: if the pilot's held-back group leaves at the same rate as the contacted group by day 45, retention aimed at notes does not work at these offers, and the money should go to fixing the billing and network causes instead.

5. Campaign engine

> Audiences are picked by written rules in code (signals.py), refreshed whenever a new notes file arrives. Every message is drafted from approved offers only, gets the legal footer added by code, and passes an automatic compliance check. 89 of 240 past campaigns would have failed that check.

The rules

Tier	Rule (only notes from the 90 days before the send)	What we send
R1 leaving now	Note says cancelling, porting or signing elsewhere	Outbound call, highest bill first; matched retention offer
R2 wavering	Note says comparing, unsure about renewal, or will leave if X	Call if capacity, else LINE/SMS; matched retention offer
R3 new	Joined less than six months ago	Welcome check-in, no paid offer
R4 unhappy, quiet	Billing or network complaint, no leave talk	Fix it: billing care or network care, no paid offer
Excluded	Line closing for a reason no offer changes; family and business plans	Nothing

One offer per customer. When more than one fits, the first match in this order wins. Retention: Device credit, Renewal discount, Data boost. Upsell: Plan upgrade, Data add-on, Roaming pass. Offer match: device problem on a premium plan → Device credit; price, rival or renewal doubt within four months of contract end → Renewal discount; price or data need and not unlimited → Data boost; billing error → fix it (not an offer); network → care call.

New campaigns

Audience | offer	Channel	Customers (full base)	Timing	Headline	Compliant
retention_R3_new_and_fragile | No offer: welcome check-in	App Push	110,890	monthly batch	Settling in with Formosa	Yes
retention_R4_unhappy_quiet | No offer: network care call	SMS	25,350	monthly batch	We are checking coverage where you are	Yes
retention_R4_unhappy_quiet | No offer: fix the bill and call back	SMS	17,550	monthly batch	We are looking at your bill	Yes
upsell_roaming_pass | Roaming pass	App Push	13,520	monthly batch	Travelling in Asia? Your first 7-day pass is free	Yes
upsell_data_add-on | Data add-on	App Push	5,460	monthly batch	20 GB more each month, first two months free	Yes
retention_R2_wavering | Renewal discount	Outbound Call	4,160	within 2 days of the note; daily notes file	Your rate plan, NT$100 a month lower	Yes
retention_R2_wavering | Data boost	Outbound Call	3,900	within 2 days of the note; daily notes file	10 GB extra every month, on us	Yes

upsell_plan_upgrade | Plan upgrade	App Push	3,510	monthly batch	Move to premium, NT$100 off for four months	Yes
retention_R1_leaving_now | Renewal discount	Outbound Call	1,690	within 2 days of the note; daily notes file	Your rate plan, NT$100 a month lower	Yes
retention_R1_leaving_now | Data boost	Outbound Call	1,300	within 2 days of the note; daily notes file	10 GB extra every month, on us	Yes
retention_R2_wavering | No offer: fix the bill and call back	Outbound Call	910	within 2 days of the note; daily notes file	We are looking at your bill	Yes
retention_R2_wavering | No offer: care call	Outbound Call	650	within 2 days of the note; daily notes file	Checking in on your service	Yes
retention_R2_wavering | No offer: network care call	Outbound Call	390	within 2 days of the note; daily notes file	We are checking coverage where you are	Yes

Example message (as sent, Outbound Call)

> Your rate plan, NT$100 a month lower
>
> Thank the customer for their time with Formosa. Acknowledge what they told us last time (they are unsure about renewing). Offer NT$100 off the monthly bill for 12 months on a new 24-month contract. If they say yes, confirm the new contract terms and end date. If no, log the reason and do not push.
>
> [Reply YES and we will set it up]

Marketer tool

A marketer types a brief and gets the audience size, the copy and a compliance result: `py make_campaign.py "renewal discount by SMS for customers who mention a rival"`

Audit of past campaigns

Problem	Campaigns
legal footer missing	68
banned claim: never drops	14
legal footer missing; banned claim: never drops	3
offer 'Renewal discount' is not approved for winback	3
legal footer missing; offer 'Renewal discount' is not approved for winback	1

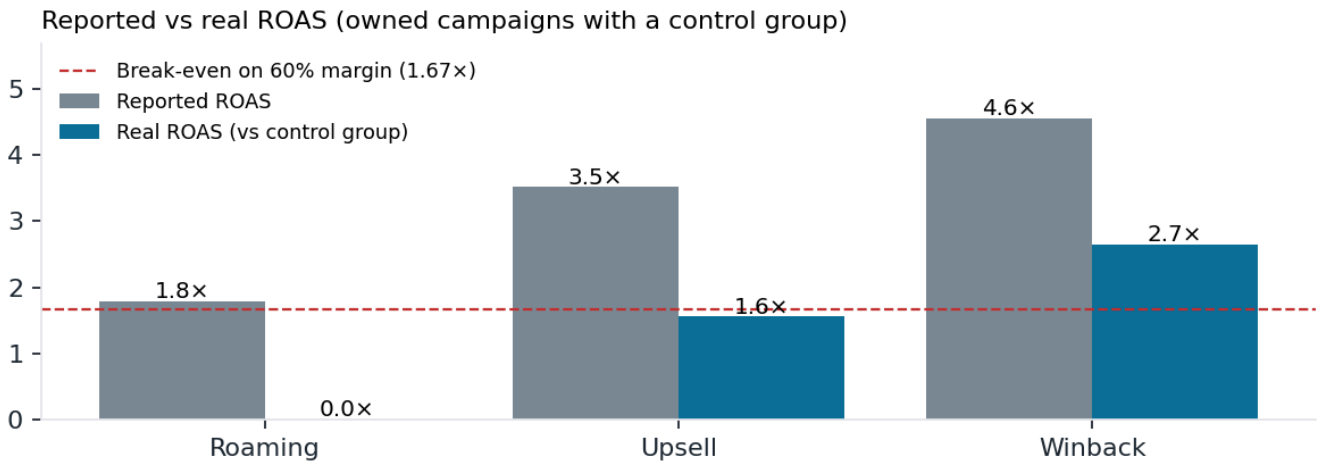
QA checklist before any send

- Offer is on the approved list for this objective; no free months, cash or bill waivers.
- Legal footer verbatim as the closing paragraph (short terms line on SMS and LINE only).
- No banned claim or close paraphrase ("never drops", "cheapest", "fastest network" unqualified...).

- Audience excludes family, business, prepaid (for retention) and "no save" lines; 20% held back at random.
- Outbound calls within 2,000 a month; prices in NT$; copy in English.

6. Measurement and the pilot

> Past owned campaigns earned less than reported: against the group held back on purpose, roaming caused nothing and upsell caused 44% of what it claimed. The pilot is built so this mistake cannot happen again: 20% held back at random from day one, success measured as the gap in churn between the two groups.

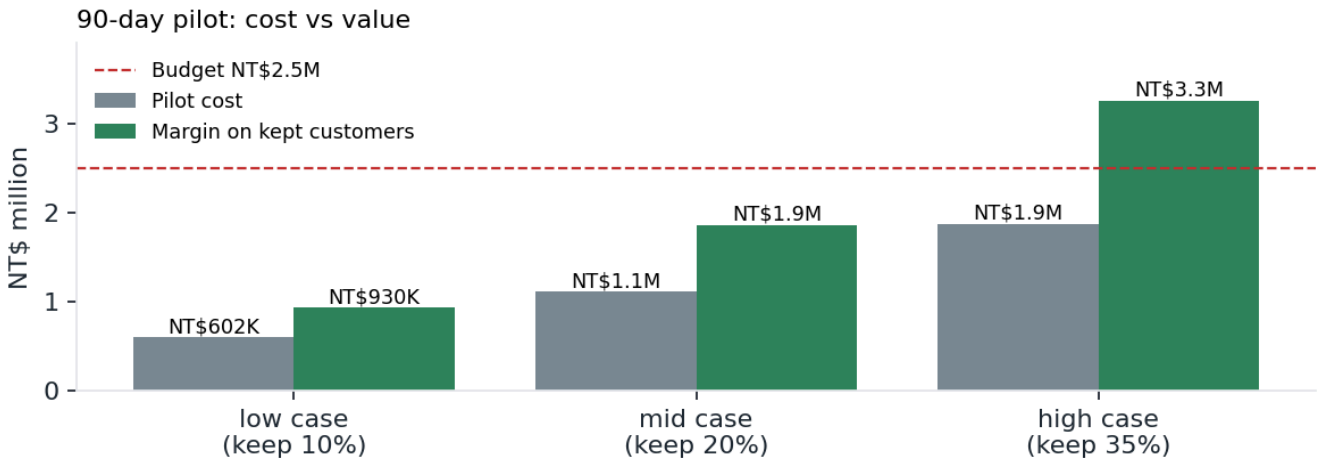


Objective	Campaigns	Spend	Reported revenue	Reported ROAS	Real ROAS	Real margin return
roaming	17	NT$2.1M	NT$3.8M	1.79×	0.00×	0.00×
upsell	39	NT$9.9M	NT$34.8M	3.51×	1.56×	0.93×
winback	12	NT$1.0M	NT$4.6M	4.55×	2.65×	1.59×

The 90-day pilot

Item	Plan
Who	Postpaid basic and premium customers in tiers R1 and R2, refreshed daily from the new notes file and contacted within two days (half of flagged leavers close their line within 48 days of the note). About 13,000 customers over 90 days at full scale.
Held back	2,600 (one in five, at random). They get nothing.
Channels	6,000 outbound calls (2,000 a month including care call-backs, highest bill first); everyone else LINE and SMS. Calls cost NT$2 per number dialled plus NT$25 per conversation; about 7.3% of dials become a conversation.
Offers	Renewal discount, Data boost or Device credit by reason; billing and network problems fixed, not discounted.
Cost	NT$1.1M expected (messages NT$24,970, offers NT$1.0M, creative NT$69,291); at most NT$1.9M. Hard cap NT$2.5M: offers stop when the cap is reached. Correcting Formosa's own billing errors is owed to the customer and sits outside the pilot budget.
Expected	Keep 513–1,796 customers; credited revenue NT$1.6M–NT$5.4M.
Break-even	Keep 2.2% of would-be leavers: churn in the contacted group 1.10 percentage points below the held-back group.

What it can prove	With 2,600 held back, chance alone moves the gap by up to ±2.1 percentage points. The pilot can reliably prove a gap of 3.1 percentage points or more, i.e. a save rate of 6.2% or more (95% confidence, 80% power). A save rate between 2.2% and 6.2% would still make money but cannot be proven in 90 days.
Day 45 check	Stop paid offers (keep care calls) if the contacted group's churn is not below the held-back group's. At day 45 noise is about ±3.0 percentage points, so this checks the direction only, not the size.
Day 90 verdict	Gap of 3.1 percentage points or more: proven, scale to the full base. Gap above zero but smaller: profitable but unproven, extend 90 days to add customers. Gap at or below zero: stop.
Measure	Churn within 60 days of contact, contacted vs held back; offer cost per kept customer; take-up by offer and channel.



On a 24-month view, at the higher churn of kept customers (12.1% a month), the pilot's margin after cost is NT$2.2M in the 20% case, against NT$750K on the six-month credit.

How many to hold back

Vivian accepts 20% or a smaller group if it can still read the result. A smaller group reaches more customers but can only prove a larger effect. We keep 20%.

Held back	Customers held back	Smallest save rate it can prove	Customers not kept (20% save rate)	Margin forgone
10%	1,300	8.3%	128	NT$106K
20%	2,600	6.2%	257	NT$211K
30%	3,900	5.4%	385	NT$317K

Risks

- **Save rate unknown.** Formosa has never run reason-matched retention. The held-back group measures it, but only effects of 6% or more can be proven in 90 days.
- **Offers taken by people who would have stayed.** Costed at 1.5 takers per customer kept.
- **Coverage.** Notes catch about 9% of leavers. Agents note almost every contact, so the rest are leavers who never contact support. Next phase: the R3 welcome check-in for new customers.
- **Call centre capacity.** 2,000 calls a month; the rest go to LINE and SMS, so the result is measured by channel.

How we used AI

Claude Opus 5.5 read the 403 distinct sentences that make up all 8,000 notes once and labelled each with a topic and a leave-intent level; it also wrote the message templates. The code only looks these up, so the results rerun offline. See 3_ai_outputs/how_we_did_it.md.